

Code : LoRa bridge project manifest (package.json)

Project reference: STN22_Thoracic belt for respiratory rate measurement

Authors: Hakkou Dounia

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File header

```
/*
Description: npm project manifest for the lora-bridge Node.js process.
Defines the package identity, module system, entry point, start script,
and all runtime dependencies needed to run server.js.

The lora-bridge runs on the Windows PC connected to the Heltec WiFi LoRa 32 V3
receiver via USB. It is a separate program from the Svelte frontend.
Launch with: npm start(or: node server.js)

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*/
```

package.json

```
{
  // Internal npm package name.
  // Must be lowercase, with no spaces or uppercase letters.
  // Identifies this project in npm registries and lock files.
  "name": "lora-bridge",

  // 1.0.0 = first stable release. Increment when making changes:
  //   patch (1.0.1) for bug fixes, minor (1.1.0) for new features,
  //   major (2.0.0) for breaking changes.
  "version": "1.0.0",

  // Declares that all .js files in this project use ECMAScript Modules (ESM).
  // This enables 'import' / 'export' statements in server.js.
  // Without this field Node.js treats .js files as CommonJS (require / module.exports)
  // and server.js would crash on its first 'import' statement.
  "type": "module",

  // Declares the main entry point of this package.
  // When another package requires/imports this package, Node.js starts here.
  // In practice, this also documents that server.js is the single entry point.
  "main": "server.js",

  // npm lifecycle scripts. Run with: npm run <name>
  // 'npm start' is a special alias that runs the 'start' script without 'run'.
  "scripts": {
    // Starts the LoRa bridge by executing server.js with the system Node.js.
    // Preferred over calling 'node server.js' directly because:
    // - ensures the script runs from the correct working directory
    // - can later be extended (e.g. 'nodemon server.js' for auto-restart)
    "start": "node server.js"
  },

  // Runtime dependencies: installed by running 'npm install' in this folder.
  // npm downloads each package into node modules/ and records exact versions
  // in package-lock.json to ensure reproducible installs on any machine.
  // The caret (^) means: accept any compatible version >= the stated one
  // within the same major version (e.g. ^13.0.0 accepts 13.0.1, 13.2.0, ...).
  "dependencies": {
    // serialport:
    // Node.js library for reading and writing to USB serial / COM ports.
    // Opens the COM port connected to the Heltec WiFi LoRa 32 V3 receiver.
    // Exposes the SerialPort class used in server.js:
  }
```

```
// const port = new SerialPort({ path: 'COM5', baudRate: 115200 })
"serialport": "^13.0.0",

// Transform stream that buffers raw bytes from the serial port and emits
// one complete UTF-8 string for each line ending in \r\n.
// This matches the format produced by the Heltec Arduino firmware:
// 'DATA:RPM:18.5\r\n' or 'LoRa TX: R:18.5\r\n'
// Used in server.js as:
// const parser = port.pipe(new ReadlineParser({ delimiter: '\r\n' }))
"@serialport/parser-readline": "^13.0.0",

// axios v1.6:
// Promise-based HTTP client for Node.js (and browsers).
// Used to send POST requests to the RespiraMed REST API:
// POST /set-patient - registers the active patient, opens a session
// POST /measurements - records one RPM measurement in the database
// POST /stop-session - closes the active session
// Preferred over the native fetch () for its cleaner error handling:
// err.response.data gives the parsed server error body directly.
"axios": "^1.6.0",

// dotenv v16:
// Reads key=value pairs from a .env file in the project root and injects
// them into process.env before any application code runs.
// Allows machine-specific settings without modifying source code:
// COM_PORT=COM3
// API_URL=https://sisteamnancy.fr/api-respiramed
// The .env file must be added to .gitignore to avoid leaking credentials.
// Called in server.js as: dotenv.config() (before any process.env read)
"dotenv": "^16.3.1"
}
}
```